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Study on characteristics of droplet transfer in low power pulsed laser-induced MAG hybrid welding with varied workpiece inclination angles

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Metrics

Outline

Highlights

Abstract

Keywords

1. Introduction

2. Material and experimental procedure

3. Results and discussion

4. Conclusions

Declaration of Competing Interest

Highlights

- Low power pulsed laser-induced MAG hybrid welding can be applied to welding at a certain workpiece inclination angle, with excellent welding stability, proving its potential for application in all position welding.
- The mechanism of D_{LA} and workpiece inclination angle on droplet transfer in low power pulsed laser-induced MAG hybrid welding was revealed.
- This work provides a theoretical basis for the popularization and application of laser-induced MAG all-position hybrid welding technology.

Abstract

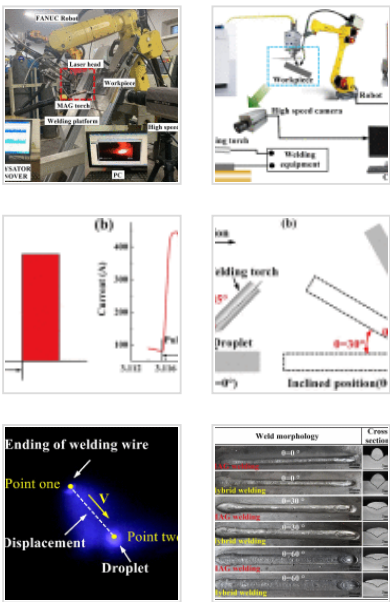
In this work, the effects of workpiece inclination angles (θ), distance between laser spot and wire tip (D_{LA}) on arc morphology, droplet transfer characteristics and stability were investigated by high-speed images (*HSI*), mathematical statistics and real-time electric signals. The coupling effect of heat source was evaluated by taking penetration as the main reference index. At the same time, the stability of the droplet transfer was evaluated based on the droplet transfer characteristics and the variation coefficient of current and voltage. The results showed that D_{LA} had significant effect on droplet transfer in laser-induced MAG hybrid welding with varied workpiece angles. At D_{LA} was 1 mm, the laser plasma compressed the arc, leading to the increase of the electromagnetic resistance of droplet transfer, which would prolong the formation and shedding time of droplet, making the droplet transfer mode change from “multi drop per pulse” to “one drop per pulse”, and improving the stability of droplet transfer. The D_{LA} affected the droplet transfer path and space instantaneous flight velocity. When D_{LA} was -1 and 1 mm, the displacement of the droplet transfer reaches the minimum and maximum, respectively. When D_{LA} was 1 mm, the droplet transfer was the most stable and maximum velocity could be obtained with varied workpiece inclination angles. This work provides a

Data availability

References

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Figures (18)



Show 12 more figures 

Tables (1)

 Table 1. Experimental parameters.

theoretical basis for the popularization and application of laser-induced MAG all-position hybrid welding technology.

Keywords

Workpiece inclination angle; D_{LA} ; Laser-induced MAG; Hybrid welding; Droplet transfer; Welding stability

[< Previous article in this issue](#)

[Next article in this issue >](#)

1. Introduction

Laser arc hybrid welding has the advantages of fast welding speed, less heat input, good gap bridging ability and strong penetration ability, and has been widely used in pressure vessels, pipelines, shipbuilding and other industries [1], [2]. Low power laser-induced arc hybrid welding technology is a kind of low energy consumption and high efficiency welding technology, which is mainly based on arc and assisted by laser. It can effectively regulate and control the arc and droplet state, improve welding stability and weld formation by utilizing laser to induce and compress arc [3].

In the actual automatic welding process, it is inevitable to weld in the non-horizontal position [4], [5]. In the hybrid welding process of varied spatial positions, due to the different stress states of liquid metal in the molten pool, when the workpiece inclination angle is large, it is easy to destabilize the molten pool, which will cause interference to arc morphology and droplet transfer, and then affect the welding stability and weld formation. Meanwhile, the distance between laser and arc (D_{LA}) is one of the crucial factors influencing the synergic effect and stability of the hybrid welding. By adjusting D_{LA} can obtain different heat source states, which has much influence on the welding process, and its influence on the laser-arc hybrid welding process has been studied by many

researchers. Hence, it is of great reference significance to investigate the effect of D_{LA} and the inclination angle on the droplet transfer and weld formation for improving the welding stability and process parameters of laser arc hybrid welding in non-horizontal position.

Liu et al. [6] studied the shape, electron temperature and electron density of mixed plasma in laser arc hybrid welding. It is found that the D_{LA} will affect the interaction between laser and arc. Ning et al. [7] applied a 3D computational fluid dynamics (CFD) based model to simulate a pulsed Nd: YAG laser TIG hybrid welding process, and found the D_{LA} would affected the shape of keyhole and thus influencing the laser energy coupling efficiency. Li et al. [8] explored the synergetic effect on the process combination of pulsed laser and pulsed MIG hybrid welding. Liu et al. [9], [10] explored the coupling-enhanced discharge phenomenon between laser and double-arcs, and indicated that the appropriate D_{LA} is beneficial for enhancing the melting efficiency of heat sources. Xu et al. [2] studied the effect of D_{LA} on energy transfer behavior in keyhole during laser-arc hybrid welding titanium alloy and found that D_{LA} greatly affects the energy transfer mode in keyhole. Gao et al. [11] investigated the effect of D_{LA} on arc characteristics, droplet transfer, and weld quality in Laser-MIG hybrid welding of 316L stainless steel. Huang et al. [12] studied the effect of D_{LA} on the gap bridging capability of hybrid welding under different butting gaps, and denoted that adjusting the D_{LA} can optimize the weld penetration and forming. According to references [13], [14], where the effect of D_{LA} on the penetration and weld geometry during laser arc hybrid welding was investigated. The electron and ion densities, electrical conductivity, and arc stability decrease with increasing D_{LA} during hybrid welding. Bunaziv et al. [15] used laser-MIG welding 5083 aluminum alloy, and indicated pores decreased with the increased D_{LA} in arc leading mode, opposite in laser leading mode. Liu et al. [16] and Moradi et al. [17] declared the D_{LA} seriously affected the laser arc hybrid process stability, and indicated that a too small D_{LA} was not conducive to process stability.

At present, there are few researches on non-horizontal position hybrid welding, and most of them focus on the several special positions such as vertical welding, overhead welding

and transverse welding. Chen et al. [18] demonstrated the droplet transfers at different positions (flat position, horizontal position, vertical position) by high speed video during CO₂ laser–MAG hybrid welding of E36 steel. The results indicated that the impact location of droplet was mainly affected by the electromagnetic force and gravity. Ma et al. [4] investigated the characteristics of droplet transfer, molten pool formation in the oscillating laser hot-wire TIG hybrid welding of 10mm thick 316LN austenitic stainless steel in the vertical position.

In summary, few researches were conducted about the effect of D_{LA} and workpiece inclination angle on the characteristics of droplet transfer in low power pulsed laser-induced MAG hybrid welding. The research on the stress state and motion characteristics of droplet transfer are still lacking. Therefore, the primary objective of this study was to investigate the effect of D_{LA} and workpiece inclination angle on the characteristics of droplet transfer in hybrid welding. The high-speed images (HSI), mathematical statistics and real-time electric signals are used to obtain the information of arc morphology, droplet transfer characteristics and stability in the process of hybrid welding with varied D_{LA} and workpiece inclination angles in real time and accurately. According to the captured image information, the force and motion characteristics of droplets in arc space are given through measurement and mathematical calculation. Based on the results, the mechanism of the effect of D_{LA} and workpiece inclination angle on the behavior of droplet transfer and the change in the forces affecting the droplet transfer were discussed.

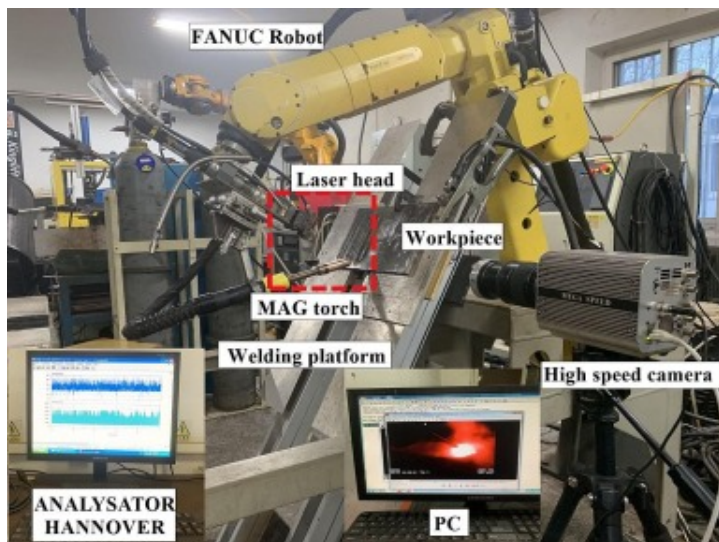
2. Material and experimental procedure

2.1. Materials

Q345 steel plates with the dimension of 200mm×100mm×6mm were used in the study. The specimen surfaces were chemically cleaned by acetone before welding to eliminate surface contamination. The type of filler wire was H08Mn2SiA (ER50-6) with diameter of 1.2mm.

2.2. Laser-induced MAG hybrid welding process

The experimental setup for the low power pulsed laser-MAG hybrid welding process was shown in Fig. 1, which mainly consists of a 1.0kW Nd:YAG solid state laser (model Trupulse 556 by TRUMPF) and a pulsed metal active gas (MAG) welding system (model Lorch S8) and a workbench with adjustable tilt angle. The Nd:YAG laser with a wavelength of 1.06mm was operated. During the experiment, the laser beam was focused using a fixed focus lens with a focal length of 150mm. By placing the sample-material top surface to the focal point ($D_f=0\text{mm}$), the spot diameter of 0.6mm was achieved. The shielding gas (20%CO₂+80%Ar) flow in the MAG welding torch was 20L/min. The welding experimental parameters are given in Table 1.



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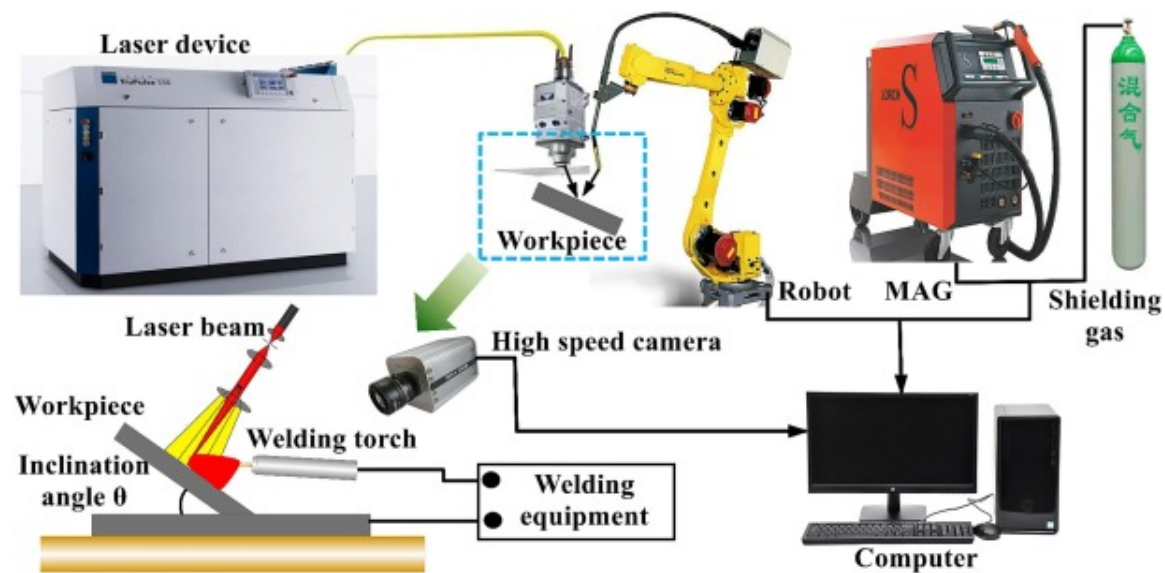
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Fig. 1. Laser-induced MAG hybrid welding experimental set-up.

Table 1. Experimental parameters.

Welding parameters	Value
Welding speed v [m/min]	0.72
Angle of welding torch α [°]	45
Defocus amount D_f [mm]	0
Extension length L [mm]	12
Wire diameter d [mm]	1.2
Average power of pulsed laser P [kW]	0.48
MAG current I [A]	200
Pulse frequency of laser f_l [Hz]	30
Pulse frequency of arc current f_a [Hz]	143
Distance between laser and arc D_{LA} [mm]	-1, 0, 1, 2, 3
Workpiece inclination angle θ [°]	0, 30, 60

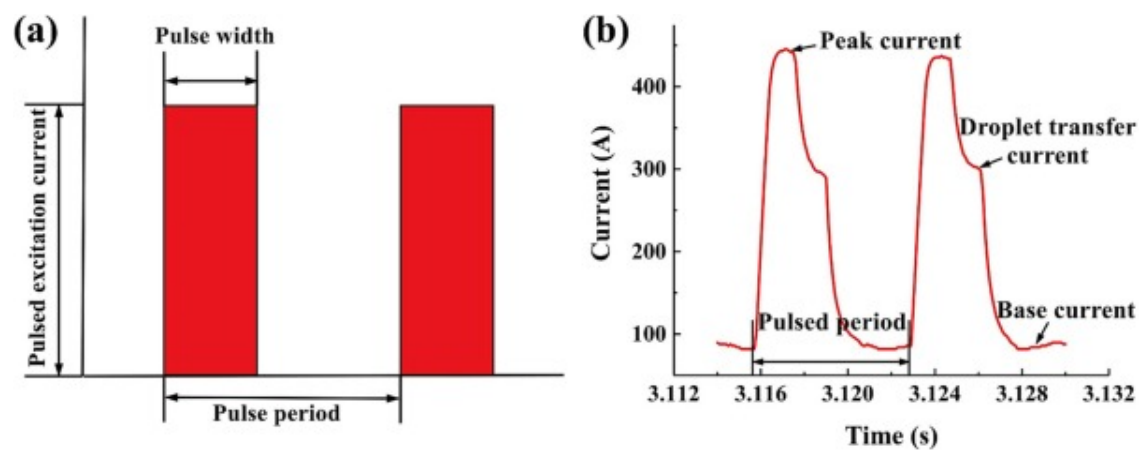
Fig. 2 illustrates the schematic diagram of a hybrid welding device. The laser was applied vertically to the surface of the base metal, the MAG welding torch was combined with the side axis of the laser head, and the angle between the laser beam and the welding torch(α) was 45°. The working mode of the laser was pulsed wave mode and MAG arc adopted DC reverse connection mode, the output of current and voltage was DC pulse mode. The pulsed waveform of laser and welding arc are shown in Fig. 3, and the phase matching mode was random matching. During the arc-leading mode hybrid welding process, the bead-on plate welding was adopted.



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Fig. 2. Device diagram of Hybrid welding with varied workpiece inclination angles.



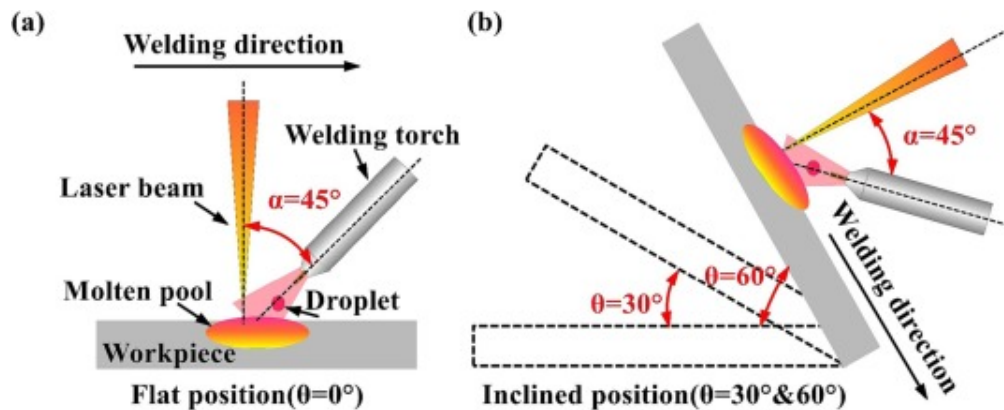
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Fig. 3. Pulsed waveform of laser and welding arc.

The visualization of arc plasma and droplet transfer was obtained by placing the camera perpendicular to the welding direction, and a MS 50K high-speed color camera (Mega Speed, Canada) with an AF Zoom-Nikkor 28–105 mm f/3.5–4.5D IF lens was used. The images were recorded using the high-speed camera at a sampling frequency of 2000 frames/s, and a narrow-band filters with the central wavelengths of 659.5 nm and was placed in front of the camera lens to filter and observe the arc plasma and droplet. To evaluate the effect of D_{LA} on the stability of droplet transfer with varied workpiece inclination angles in hybrid welding, the electrical parameters were also collected by ANALYSATOR HANNOVER (AH19) with a sampling rate of 2 kHz during hybrid welding.

The welding schematic diagram with varied workpiece inclinations was shown in Fig. 4. The relative position relationship between the laser and the arc remained unchanged as the workpiece inclination changed. The inclination angle of the workpiece changed from 0° to 30° and then to 60° .

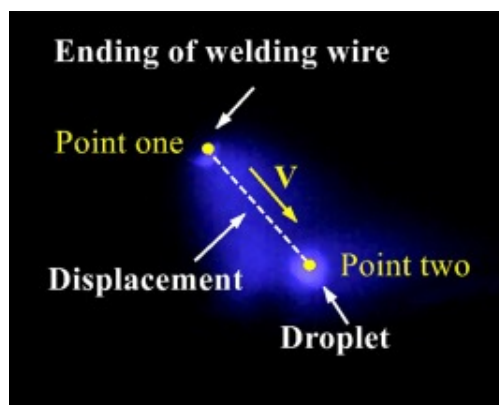


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Fig. 4. Welding schematic diagram with varied workpiece inclinations.

As schematically shown in Fig. 5, the motion characteristics of the droplet of hybrid welding, defined by its displacement and velocity, was suitably measured by properly computerized scaling technique, which applied on a number of photographs of droplet falling process. In order to maintain uniformity in comparable measurement, the displacement of droplet transfer was measured with the help of the Photoshop software. The detachment of droplet from the wire end was continuous, which was taken by high speed camera in consecutive frames of video-graphs. The moving distance of the droplet was determined by measuring the distance between the tip of filler wire and the droplet. The curves of velocity curve of droplet were obtained by calculation of the time displacement curves.



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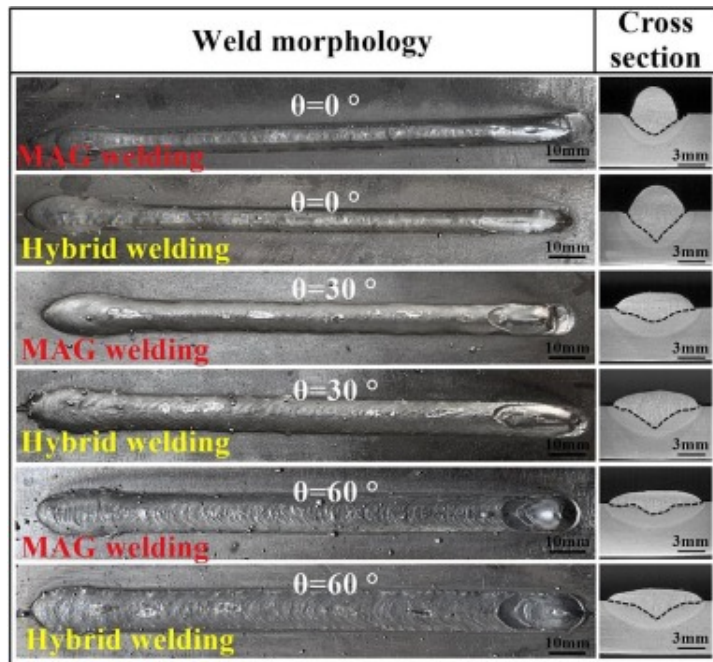
Fig. 5. Sketch map of displacement measurement of droplet in arc space.

3. Results and discussion

3.1. Analysis of weld forming

Fig. 6 shows the typical morphology of weld and section obtained with varied workpiece inclination angles with laser power of 480W, current of 200 A, welding speed of

0.72m/min. It can be seen that the addition of laser can significantly improve the penetration. With the increase of workpiece inclination, the weld penetration and reinforcement decrease, and the weld width increases. An explanation could be as follows. As workpiece inclination angle increasing, the value of gravity component gradually increases along the welding direction, and the resultant force of liquid metal becomes along the welding direction. The liquid metal in the molten pool is prone to flow phenomenon, and the trend of liquid metal flowing towards the front end of the molten pool under gravity intensifies. A large amount of liquid metal is deposited in the center of the molten pool, which forms upwelling and circulation under the action of arc force, etc., and the increase of heat leads to the outward expansion of the molten pool, the increase of width and the decrease of residual height.

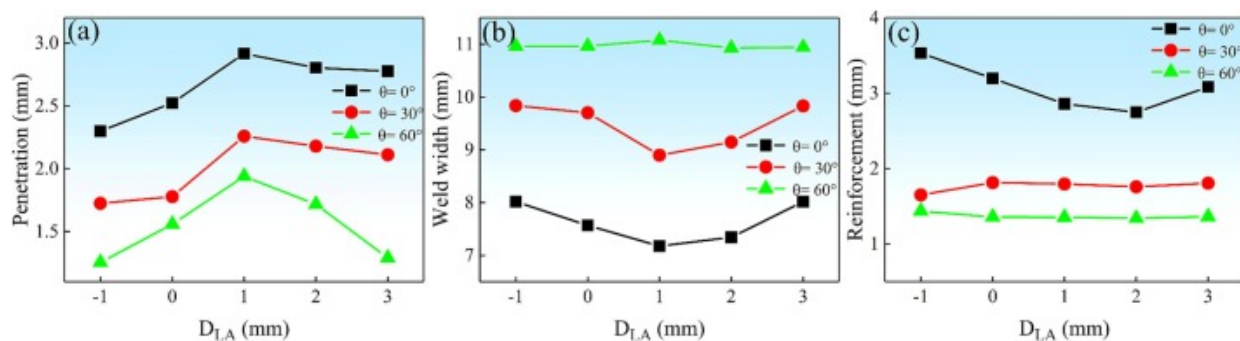


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Fig. 6. Typical morphology and cross sections of weld bead with varied workpiece inclinations.

Fig. 7 presents the effect of D_{LA} on penetration, weld width and reinforcement with varied workpiece inclination angles. From Fig. 7(a), it can be seen that with the increasing of θ from 0 to 60°, the penetration and reinforcement of weld decreased correspondingly. Concurrently, under the same workpiece inclinations, the penetration of weld increased first and then decreased with the increased of D_{LA} . When D_{LA} was 1 mm, the penetration reached the maximum. In Fig. 7(b), when the θ were 0° and 30°, along with the increased of D_{LA} , the weld width first decreased and then increased, and the minimal weld width was obtained at D_{LA} was 1 mm. When the θ increased to 60°, D_{LA} had little effect on the melting width and the weld width was basically stable. When the θ was 0°, the reinforcement decreased first and then increased with the increase of D_{LA} , which can be observed from Fig. 7(c). It achieved the minimum values when D_{LA} was 2 mm. When the θ were 30° and 60°, D_{LA} had little influence on the reinforcement, and the fluctuation of welded reinforcement was very small within a certain range.



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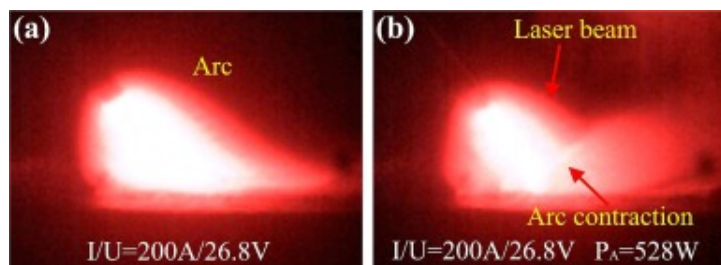
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Fig. 7. Effect of D_{LA} on weld characteristics with varied workpiece inclinations. (a) Penetration; (b) Width; (c) Reinforcement.

3.2. Analysis of D_{LA} on arc shape with varied workpiece inclinations

Fig. 8 shows the different arc shapes between MAG welding and laser-induced MAG hybrid welding. D_{LA} plays a very significant role in the coupling regulation of laser and

arc, which was embodied in the morphological changes of arc (expansion, deflection and contraction). In the traditional MAG welding, the arc shape was relatively divergent, nevertheless, in laser-MAG hybrid welding, the addition of laser led to the contraction of arc shape and the arc deflected towards the laser, which can be observed from Fig. 8(b).

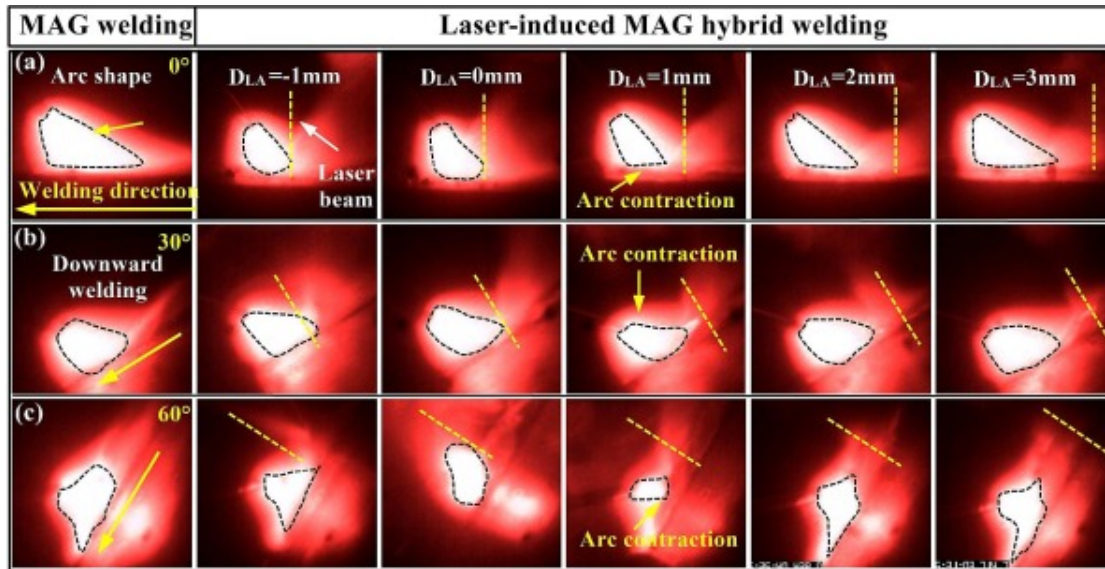


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Fig. 8. The Arc plasma shapes with varied welding.

Fig. 9 demonstrates the influence of D_{LA} on the arc shape with varied inclination angle of workpiece. When the θ was 0° , the influence of D_{LA} on the arc shape was shown in Fig. 9(a). When D_{LA} was appropriate (D_{LA} was 1 mm), the pulsed laser acted on the arc column area instantaneously, it could be seen that due to the induced effect of laser on arc, the area of arc column decreased as a whole and the arc stiffness increased. In Fig. 9(b), when the θ was 30° , the downward welding was adopted, and a smaller D_{LA} was conducive to heat source coupling. When D_{LA} was 1 mm, the induced compression effect of pulsed laser on arc was the best, the arc shrunk and deflected obviously, a stable discharge channel was generated between arc and plate. The conductive area in the center of arc column was compressed and the contact area between arc and plate was reduced to the minimum, which was more contributive to the heat and mass transfer between arc and plate.



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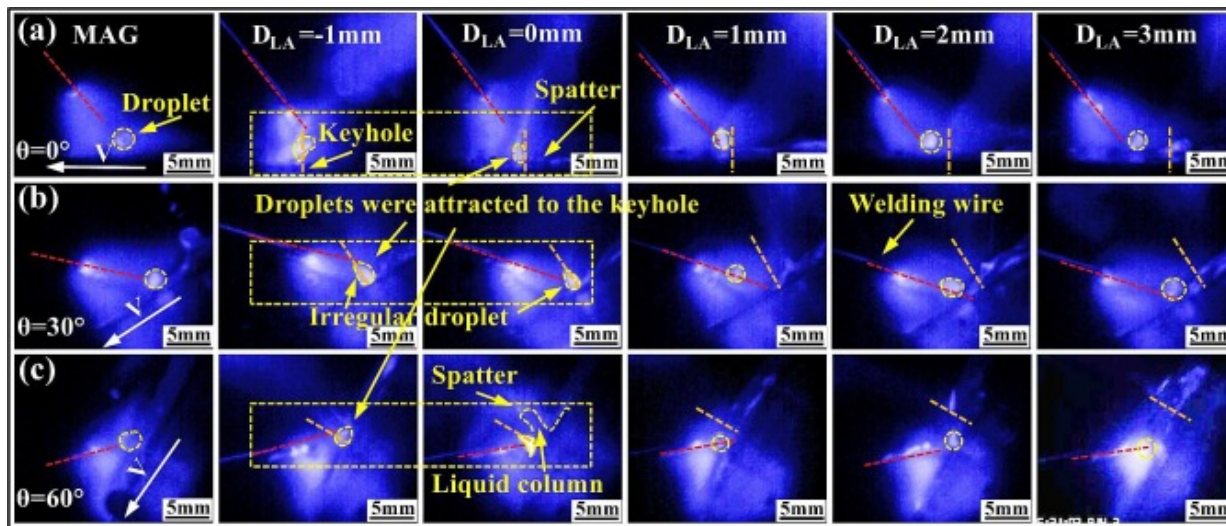
Fig. 9. Arc plasma shapes with varied workpiece inclinations.

As depicted in Fig. 9(c), when the θ was 60° , the downward flow of the molten pool was intensified under its own gravity, which interfered with the stability of arc. The shape of arc was particularly unstable and D_{LA} was more sensitive to the coupling state. When the D_{LA} was excessive, the front of the molten pool rose due to the action of gravity, resulting in more molten metal between the laser and the arc. Because the interaction area between laser and arc was not in a regular molten pool plane, the distance between arc plasma and laser plasma was relatively far, which was not conducive to the coupling between them, and the plasma between them was weakly connected. With the decrease of D_{LA} , the relative distance between the laser irradiation area and the arc area decreased, and the laser and arc gradually began to couple. When D_{LA} was 1 mm, the arc was compressed to the minimum, the arc stiffness was large and the arc length was shortened.

3.3. Analysis of D_{LA} on droplet transfer with varied workpiece inclinations.

3.3.1. Morphological analysis of droplet transfer

Fig. 10 shows the droplet transfer with similar spatial dimensions with varied workpiece inclination. When D_{LA} was in the range of -1 and 0 mm, the keyhole generated by pulse laser would affect the droplet transfer, and the molten droplets were forcibly attracted to the keyhole and deviated from its original transfer path in all workpiece inclinations. The droplet would fly directly above the keyhole and be hit by the laser beam, resulting in splashing and interfering with the normal flying of the droplet into the molten pool. The impact of droplets on the keyhole would accelerate the closure and collapse of keyhole, decreased the existence time of keyhole, which was detrimental to the transfer of arc heat to the bottom of molten pool.



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Fig. 10. Droplet transfer with similar spatial dimensions under varied workpiece inclinations.

It could explain in this way: when D_{LA} was small, the radiation heat effect of laser plasma on the end of welding wire was intensified. The end of welding wire would not only be heated and melted by a large amount of Joule heat of arc during the peak current period, but also the heat radiation effect of laser energy would provide a large amount of energy for the melting of welding wire end, resulting in easier melting of welding wire end and formation of droplets. Meanwhile, the metal vapor reaction force hindered the droplet, the inner bottom of droplet would be affected by the upward electromagnetic force, and the top of droplet would be affected by the downward electromagnetic force, resulting droplet would be squeezed. When it fell off, the shape was irregular, which was easy to cause a large amount of splash. Owing to it was close to the keyhole, the cathode spot formed by the laser keyhole had a stronger effect, making the deflection angle between the electromagnetic force and the axial direction of the welding wire increased. The droplets were easier to fly to the keyhole rather than directly into the molten pool.

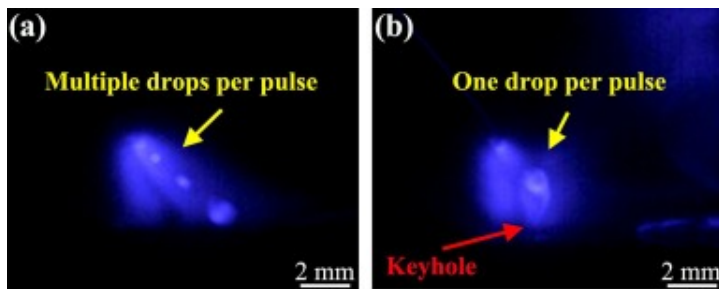
When D_{LA} was 1 mm, the thermal radiation effect of laser on the end of welding wire, the reaction force of metal vapor sprayed from keyhole on droplet and the charge flow effect caused by the formation of conductive channel and laser plasma would be weakened. Due to the arc plasma action area was far from the keyhole, the charge flow effect of the arc was very weak, while the plasma flow effect was dominant, and the deflection angle between the electromagnetic force and the axial direction of the welding wire decreased. The necking effect at the end of welding wire was strong, so the droplet was easier to fall off and separate towards the axis of the welding wire. The shape of droplet was regular and entered the molten pool regularly, would not hit the keyhole, and was little affected by the metal vapor caused by the laser. As D_{LA} continued to increase to 3 mm, the laser keyhole was far away from the arc action area. There was almost no coupling between laser and arc, and the droplet transfer of hybrid welding was the same as that of arc welding.

In summary, the effect of D_{LA} on droplet transfer was basically the same with varied workpiece inclinations. However, with the increase of the workpiece angle, the molten pool was affected by its own gravity, which slowed down the backward flow of the

molten pool, resulting in the accumulation of molten metal in front of the molten pool, and the unstable liquid column would appear, which interfered with the coupling effect of laser and arc, and thus affected the normal droplet transfer.

3.3.2. Motion characteristics of droplets

Fig. 11 presents the droplet transfer of MAG welding and hybrid welding. In arc welding, the average welding current was 200 A, as the pulse current changed from base value to peak value, droplet was produced at the tip of welding wire and accumulated continuously. When the pulse welding current reached the peak value, the tip of welding wire shrunk, and the electromagnetic force generated by the current would be greater than the surface tension, so the droplet would separate from the tip of welding wire. At this time, the droplet size was large and accompanied by small droplets, which was not conducive to the welding stability, as depicted in Fig. 11(a).



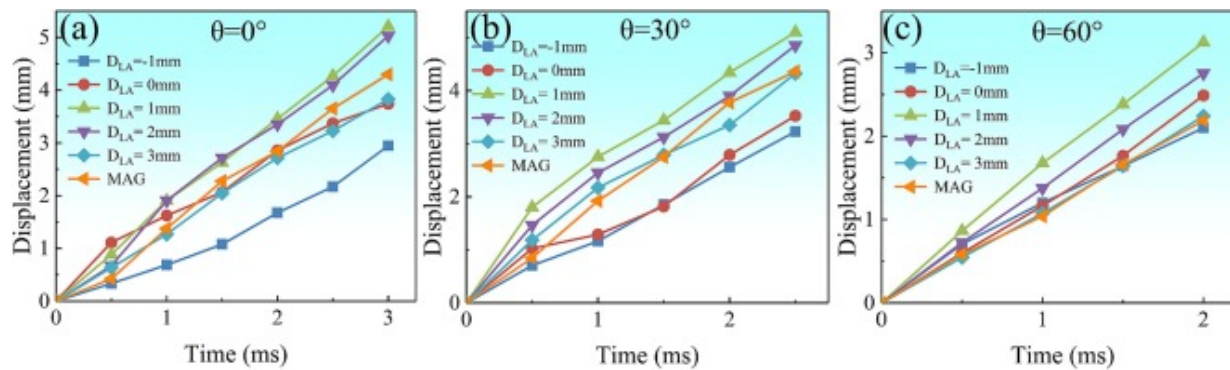
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Fig. 11. Droplet transfer mode of MAG welding and hybrid welding.

In hybrid welding, the pulse laser acted on the molten pool to produce keyhole, the high-temperature metallic vapor generated by the keyhole would produce upward vapor pressure, which reduced the droplet transfer frequency and refined the droplet size [19]. It was conducive to the smooth transfer of droplets to the molten pool, as shown in Fig. 11 (b).

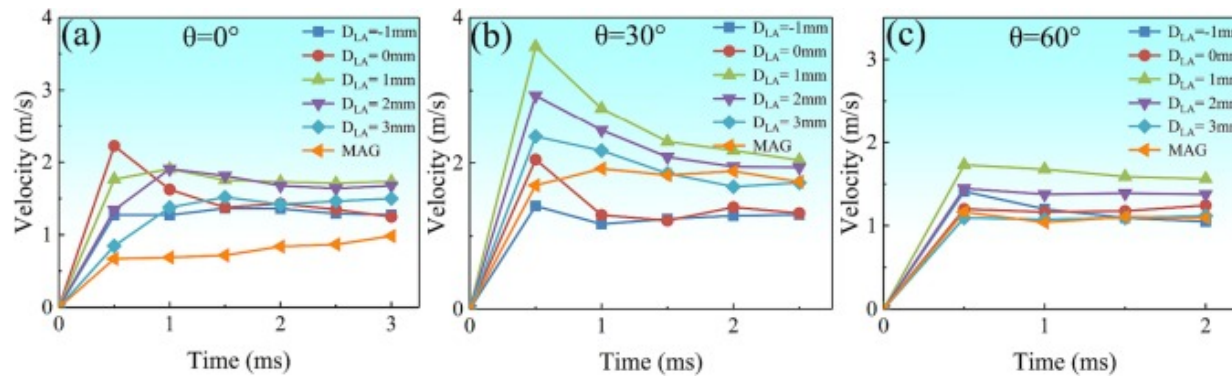
Figs. 12 and 13 show the displacement and velocity curves of droplet in the arc space with varied workpiece inclinations, respectively. As shown in Fig. 12, the droplet displacement was calculated from the detachment of the welding wire. During the flat welding (at θ was 0°), the effect of D_{LA} on droplet transfer was shown in Fig. 12 (a). The droplet transfer displacement increased first and then decreased with the increased of D_{LA} . When the D_{LA} was -1 mm, and the high-speed metal steam jet from the keyhole interferes with the normal droplet transfer, making it deviate from the axis of the welding wire and reduce the flight path. The minimum droplet transfer displacement was 2.95 mm. When D_{LA} was 0 mm, the flight path of the droplet did not deflect. When D_{LA} was 1 mm, the transfer path deflected towards the laser side, and the droplet displacement increased to a maximum of 5.2 mm. When D_{LA} was 2 mm, the phenomenon of laser induced and compressed arc still existed, arc shape shrunk slightly, and the droplet transfer displacement was 5.03 mm. When D_{LA} continued to increase to 3 mm, there was no coupling effect between laser and arc, and the heating effect of laser on welding wire and the force of keyhole metal vapor on droplet were weakened. The droplet transfer displacement reduced to 3.83 mm.



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Fig. 12. Displacement of droplets with different D_{LA} under varied workpiece inclinations.



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Fig. 13. Velocity of droplets with different D_{LA} under varied workpiece inclinations.

As demonstrated in Fig. 12(b) and (c), when the θ was 30° and 60° , the rule of droplet transfer was similar to that at 0° . As the θ increased, the droplet transfer displacement decreased. The molten metal flowed to the front of the molten pool under its own gravity, making more metal in the front of the molten pool. The distance between the end of the welding wire and the surface of the molten pool was shortened, and the transfer displacement of the drop in the arc space was reduced.

Fig. 13 shows the instantaneous velocity of droplet transfer at different times in the droplet transfer process with varied workpiece inclinations. In Fig. 13(a), the θ was 0° , and when D_{LA} was -1 mm, the velocity of droplet transfer was 1.27 m/s at 0.5 ms after the droplet was released from the welding wire, and then the velocity did not increase significantly. When D_{LA} increased to 0 mm, the droplet transfer velocity at 0.5 ms increased to 2.22 m/s and then decreased rapidly. The metal vapor generated by laser keyhole eruption would hinder the droplet transfer, which was most obvious when D_{LA} was 0 mm. When D_{LA} in the range of 1 and 2 mm, the coupling effect of laser and arc was conducive to droplet transfer, and the velocity of droplet transfer was faster. With the D_{LA} increased to 3 mm, the laser keyhole kept moving away from the droplet, the velocity of

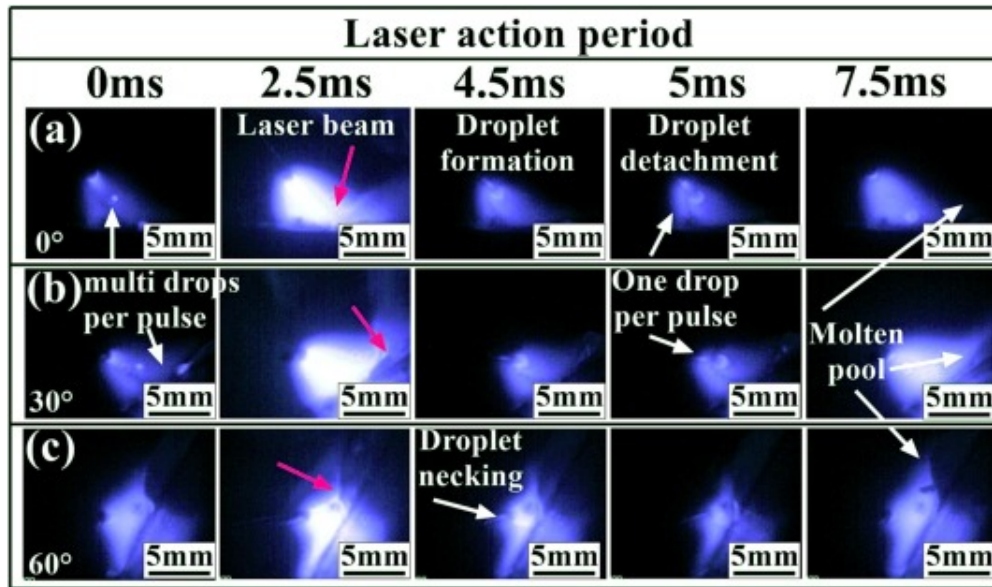
droplet transfer decreased to 0.84m/s at 0.5ms, which was similar to the velocity of droplet transfer in MAG welding.

In Fig. 13(b) and Fig. 13(c), when the θ was 30° and 60°, the effect of D_{LA} on the velocity of droplet transfer was similar to that at 0°. Meanwhile, with the increased of θ , the component force of the droplet gravity on the axis direction of the welding wire would change from promoting the droplet transfer to hindering the droplet transfer, resulting in a decrease in the velocity of the droplet after it was released from the welding wire, which was most obvious when the θ was 60°.

The above results show that: first of all, D_{LA} had a great influence on droplet transfer in laser-induced MAG hybrid welding with varied workpiece inclinations. Too small D_{LA} would make the reaction force of metal vapor hinder droplet transfer and reduced the transfer speed. Meanwhile, when D_{LA} was too large, the coupling effect of heat source was weakened, and the velocity of droplet transfer also decreased. Secondly, compared with MAG welding, the optimized hybrid heat source achieved perfect coupling effect, promoted the droplet transfer and increased the velocity of droplet transfer.

Figs. 14 and 15 shows the droplet transfer cycle during laser action period and no laser action period with varied workpiece inclinations when the D_{LA} was 1 mm, respectively. When the laser pulse acted on the arc, the laser and the arc were coupled to achieve one pulse one droplet transfer, as illustrated in Fig. 14(a). The droplet generation time was 4.5ms. After the droplet necked away from the end of the welding wire, it flew in the arc space for 3ms, and then struck the molten pool. In Fig. 15(a), when there was no laser action, the droplet generation time was 4ms, and the droplet transfer was a mixed transfer mode, including multi pulse one drop and one pulse one drop. In Fig. 14(b)-(c) and Fig. 15(b)-(c), the droplet transfer at workpiece inclinations of 30° and 60° was similar to that at 0°. The addition of laser could also extend the droplet generation time and contribute to achieving a stable transfer of one droplet per pulse. When the θ was 60°, the molten pool flowed significantly under the action of gravity, and the time required for the droplet to fly to the molten pool after separating from the welding wire

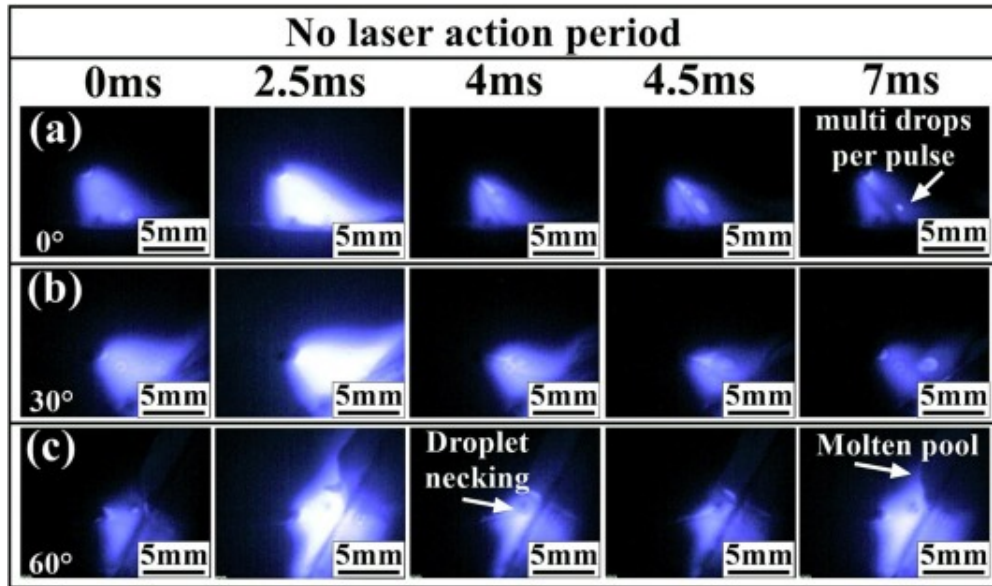
was reduced to 2.5ms. Although the droplet generation time under laser irradiation was still 4ms, the necking position of the droplet was prolonged and the droplet was elongated, indicating that it was difficult for the droplet to detach at this time.



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Fig. 14. Droplet transfer cycle during laser action period with varied workpiece inclinations.



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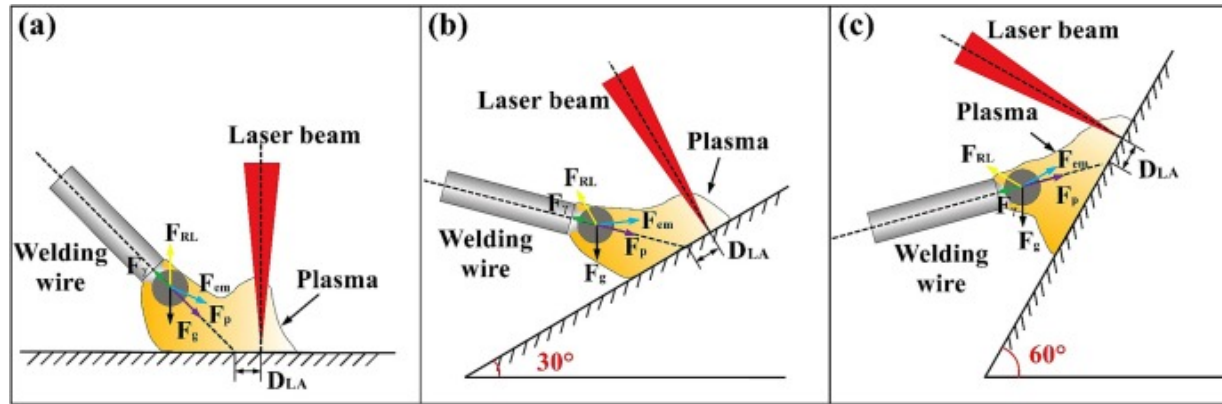
Fig. 15. Droplet transfer cycle during no laser action period with varied workpiece inclinations.

To sum up, the coupling effect of laser and arc made the formation time of droplet lengthen, which was conducive to achieving a stable pulse drop transfer. At the same time, as the inclination angle of the workpiece increased, the difficulty of droplet transfer increased.

3.3.3. Force analysis of droplet in arc space

In the laser-induced MAG hybrid welding process with varied workpiece inclinations, the force on the droplet was the main factor affecting the droplet transfer. The droplet was mainly affected by the following forces in the process of forming, falling and entering the molten pool: droplet gravity F_g , plasma flow force F_p , electromagnetic pinch force F_{em} and surface tension F_γ and laser induced steam recoil pressure F_{RL} , as illustrated in Fig. 16. The

gravity F_g , electromagnetic pinch force F_{em} and plasma flow force F_p promote the droplet transfer, and the surface tension F_γ , the laser induced steam recoil pressure F_{RL} acts as an obstacle.



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Fig. 16. Force analysis of droplet with varied workpiece inclinations.

In the arc space, the F_g can be expressed by Eq. (1)[19],

$$F_g = 4\pi r_d^3 \rho g / 3 \quad (1)$$

where r is the droplet radius, ρ is the droplet density and g is the gravitational acceleration. The F_g is directly proportional to r . When pulsed laser was applied, the droplet transfer frequency decreased, the droplet size decreased and its gravity decreased. As the inclination angle of the workpiece increased, the component force of the droplet's gravity along the welding wire axis gradually decreased, and when the θ was 0° and 30° , its component force promoted the droplet transfer. When the θ was 60° , the component force hindered the droplet transfer, as presented in Fig. 16(c).

In the process of droplet shedding, the plasma flow in the arc space flows rapidly from the end of the welding wire to the molten pool, which is conducive to droplet transfer.

The plasma flow force F_p can be represented by Eq. (2)[1],

$$F_p = \frac{C_d A_p \rho_f v_f^2}{2} \quad (2)$$

where C_d is the plasma flow coefficient ($C_d=0.45$), A_p is the action area of plasma flow, and v_f is the velocity of plasma flow, ρ_f is the density of plasma flow. According to the research of [19], [20], With the coupling effect between laser and arc, the introduction of the laser obviously compresses arc, the arc shape shrinks, especially the arc near the weld pool produces more intense contraction. Therefore, the A_p decreases. The cross-sectional of the arc column decreases, which increases the current density of the arc and the v_f . As the peak arc pressure increases, the F_p acting on the droplet increases.

Electromagnetic contraction force F_{em} is an important force affecting droplet transfer, which is the macroscopic reflection of the current in the droplet to the Lorentz force in the magnetic field. The F_{em} can be expressed by the parameters of the current in the droplet, as shown in Eq. (3)[21],

$$F_{em} = \frac{\mu_0 I^2}{4\pi} \left(\ln \frac{r_d \sin \theta}{r_w} - \frac{1}{4} - \frac{1}{1 - \cos \theta} + \frac{2}{(1 + \cos \theta)^2} \ln \frac{2}{1 + \cos \theta} \right) \quad (3)$$

where I is welding current, r_d is droplet radius, r_w is welding wire radius, θ is the angle of arc coverage, and μ_0 is the permeability of free space ($4\pi \times 10^{-7} \text{H} \cdot \text{m}^{-1}$). The shape of arc and current distribution directly determine the magnitude and direction of the F_{em} . When the pulse laser is coupled with the arc, the arc shrinks and deflects, and θ decrease, the arc shape is in the contraction form, the current distribution will appear obvious convergence state, and the F_{em} decreases.

The surface tension F_γ plays a vital role in promoting the molten metal adhesion to the tail of welding wire, and hinders the transfer of droplets. It could be expressed as follows [22],

$$F_\gamma = 2\pi R \gamma \quad (4)$$

where R is the radius of welding wire, γ is the surface tension coefficient and $\gamma \propto 1/T$, T is the temperature of droplet. When the laser was coupled with arc, the laser absorption efficiency was improved, and produced strong metal vapor to irradiate the droplet and the γ of droplet reduced. Hence, the F_γ reduced, which was more conducive to the refinement of droplet and promoted the transfer of droplet.

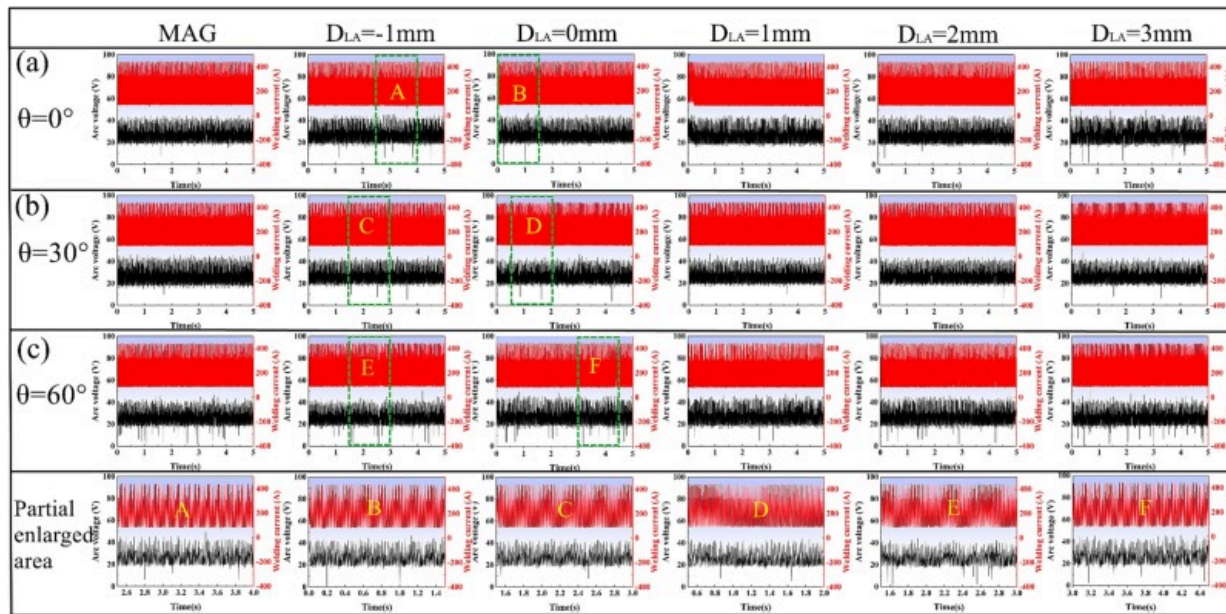
During laser-induced MAG hybrid welding, the pulsed laser beam acts on the material surface instantaneously, and the solid metal is heated, melted and vaporized rapidly to form a laser keyhole, and a large amount of metal steam is ejected from the laser keyhole, which produce an upward impact force on the droplet and hinder the droplet transfer. The upward impact force is expressed by the vapor recoil F_{RL} [23],

$$F_{RL} = \left\{ \frac{1}{4\pi R_h^2} C_d A_p \rho_m^2 v_0^2 \frac{N_a k_B T_s^{\frac{3}{2}}}{M_a B_0} \exp\left(\frac{-M_a L_v}{N_a k_B T_s}\right) \exp\left(\frac{-D_{LA}^2}{2R_h^2}\right) \right\} (D_{LA} \leq R_h) \quad (5)$$

where R_h is the metal vapor distribution parameter, A_p is the plasma action surface, and C_d is the plasma flow coefficient, ρ_m is the plasma density and V_0 is the constant ($3.4 \times 10^2 \text{m/s}$), N_a is Avogadro constant, k_B is Boltzmann constant, T_s is the surface temperature of molten metal, and D_{LA} is the heat source distance between laser and arc. It can be seen from Eq. (5) that the effect of F_{RL} on droplet transfer changes with the change of D_{LA} .

3.4. Electrical signal analysis of droplet transfer stability.

The waveform of welding current and voltage is an important symbol reflecting the stability of welding process and the droplet transfer process. To characterize the stability of D_{LA} on droplet transfer during welding with varied workpiece angles, typical electrical signal waveforms of welding current (I) and voltage (U) were used, as shown in Fig. 17. The red and black waveform represented the welding current and welding voltage, respectively.



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Fig. 17. The voltage and current waveforms of different D_{LA} with varied workpiece inclinations.

When the θ was 0° and D_{LA} was in the range of -1 and 0 mm, the laser directly impinged on the droplet due to the enhanced shielding and absorption of the arc plasma, which interfered with the normal droplet transfer. The current waveform fluctuated obviously, and the voltage waveform dropped sharply to 0 V, which indicated that the welding process was unstable at this time and some droplet transfer instability existed, as shown in the region(A)~(B) of Fig. 17(a) and their enlarged image. When D_{LA} was 1 mm, the waveform of current and voltage changed regularly, and the waveform of electric signal tended to be stable. The voltage waveform did not drop sharply to 0 V, that was, no short circuit occurred, indicating that the droplet transfer was stable, as shown in Fig. 17(a). This was the result of the coupling effect between laser and arc. First, the laser induced and compressed the arc, which enhanced the arc stability. Secondly, based on the

principle of minimum piezoelectricity [24], the electrons natural path mainly travels along the laser plasma and points to the workpiece. The laser induced and compressed the arc, resulting in changes in the conductive channel between the welding wire and the plate, as well as changes in the electromagnetic force acting on the droplet. The flight path of the droplet shifted to a certain extent, and the flight distance increased during which the droplet's influence on the molten pool was weakened. With the continuous increased of D_{LA} , the fluctuation of current and voltage waveform became worse, the effect of laser on the droplet transfer became worse, arc plasma and laser plasma separated from each other, and the overall stability of droplet transfer became worse.

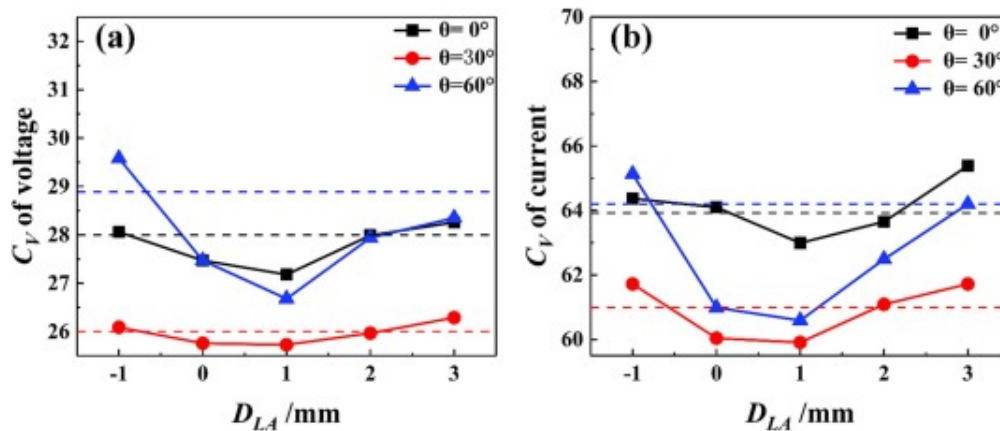
When the θ was 30° and 60° , the influence of D_{LA} on the stability of droplet transfer was similar to that at 0° , as shown in Fig. 17 (b) and Fig. 17 (c). When the θ was 30° , the welding current and voltage waveform was the most stable. The reason was that when the θ was 30° , the molten pool flowed downward along the inclined plane under the action of its own gravity, which slowed down the backward flow trend of the molten pool. To a certain extent, it was conducive to wetting and spreading the weld toe of the molten pool on both sides, which weakened the dynamic fluctuation of the molten pool, facilitated the smooth and stable droplet transfer to the molten pool, and improved the weld formation. When the θ was 60° , the dynamic characteristics of welding voltage waveform fluctuated seriously, short-circuit phenomenon occurred for many times, and the stability was poor, as indicated in Fig. 17(c). The reason was that the molten pool under its own gravity made the liquid metal at the front of the molten pool too much, and the molten pool fluctuated greatly, which seriously interfered with the normal droplet transfer.

To further investigate the effect of D_{LA} on the stability of droplet transfer in the welding process with varied workpiece inclinations, the variation coefficients of voltage and current was analysed, as depicted in Fig. 18. Coefficient of variation (C_V) can be calculated

by Eq. (6)[22],

$$C_V = \frac{\sigma}{\mu} \frac{\sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \mu)^2}}{\frac{1}{n} \sum_{i=1}^n x_i} \quad (6)$$

where σ and μ are the standard deviation and mean value, respectively. N and x_i are totals and continuous random variables. In the welding process, the C_V of current and voltage was not affected by welding parameters, which objectively described the fluctuation degree of welding process. The smaller the C_V , the smaller the fluctuation, the better the arc stability, the more stable the welding process.



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Fig. 18. Effect of D_{LA} on coefficient of variation (C_V) with varied workpiece inclinations: (a) Voltage; (b). Current.

Fig. 18 illustrates the C_V curves of voltage and current with different D_{LA} under varied θ . The influence trend of D_{LA} on the stability of droplet transfer with varied θ was similar. With the increase of D_{LA} , the C_V of current and voltage first decreased and then increased. The minimum C_V appeared when D_{LA} was 1 mm, and the C_V was lower when the θ was 30° , indicating that the θ also had an effect on the stability of droplet transfer. Under the

experimental conditions, when the θ was 30° , the overall stability of droplet transfer was the optimal.

4. Conclusions

The characteristics of droplet transfer in low power pulsed laser-induced MAG hybrid welding with varied workpiece inclination angles were investigated. The main conclusions can be summarized as follows:

(1) Laser-induced MAG hybrid welding was suitable for welding with varied workpiece inclinations and D_{LA} has significant effect on droplet transfer. At D_{LA} was 1 mm, the laser plasma compresses the arc, leading to the increase of the electromagnetic resistance of droplet transfer, which will prolong the formation and shedding time of droplet, making the droplet transfer mode change from “multi drop per pulse” to “one drop per pulse”, and improving the droplet transfer stability.

(2) The different D_{LA} affect the droplet transfer path and space instantaneous flight velocity. When D_{LA} was -1 mm, the droplet transfer displacement is minimum, and when D_{LA} was 1 mm, the droplet transfer displacement is maximum. On the whole, the droplet transfer displacement is minimum when the θ was 60° . As the D_{LA} increases, the space flight velocity of the droplet increases first and then decreases. When D_{LA} is 1 mm, the droplet can obtain the maximum velocity with varied workpiece angles.

(3) As the inclination angle of the workpiece increases, the component force of the droplet gravity in the direction of the welding wire axis changes from promoting droplet transfer to hindering droplet transfer. When θ was 60° , the necking part of the droplet was prolonged. It is not conducive to the realization of “one drop per pulse”, and the stability of droplet transfer becomes poor.

(4) From electrical signals and coefficients of variation (C_V) of current and voltage, with D_{LA} was 1 mm, stable droplet transfer can be obtained with varied workpiece inclination angles. The stability of droplet transfer was the best when θ was 30° .

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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